

- **Recurrent states** are dynamic activations that are fed back into the next update. Quantizing them changes not only one layer output but also the numerical state from which later predictions are computed.

The nonlinear function f is an operation rather than a stored tensor. Its input and output may be quantized, while the operation itself may be evaluated through integer arithmetic, a lookup table, a polynomial approximation, or a floating-point routine. Consequently, the label “quantized network” is incomplete unless the precision of weights, biases, activations, states, accumulators, and arithmetic is stated explicitly.

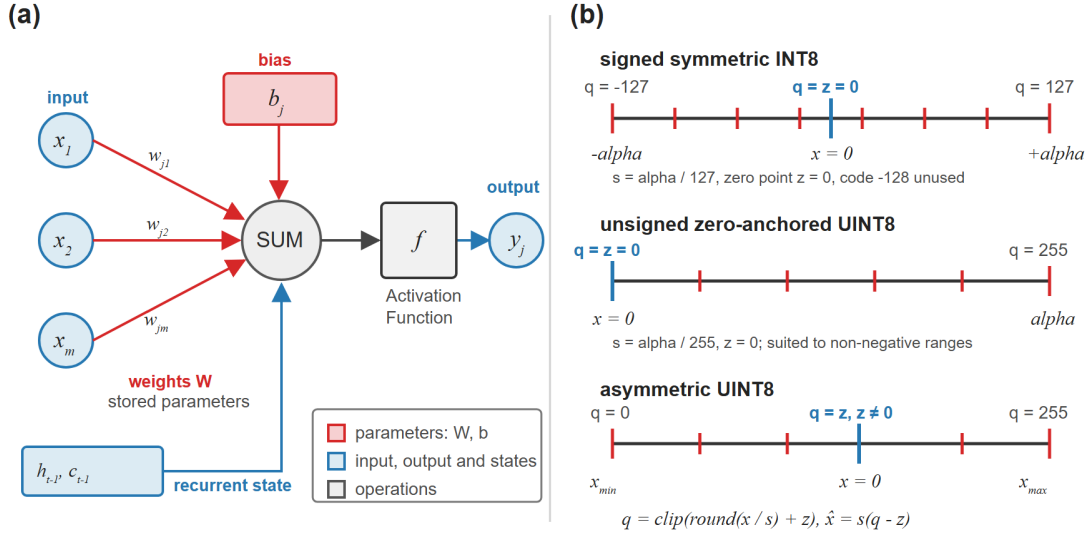


Figure 3.7: Quantizable quantities and representative uniform 8-bit grids. Panel (a) distinguishes stored weights and biases, dynamic inputs, outputs and recurrent states, and the summation and activation operations of a neural computation. From top to bottom, panel (b) shows signed symmetric INT8 with $z = 0$, unsigned zero-anchored UINT8 with $z = 0$ for non-negative values, and asymmetric UINT8 with $z \neq 0$. Labels above each grid denote integer codes q , whereas labels below denote their reconstructed real values x . The signed symmetric example uses the exactly balanced code range -127 to 127 and leaves the INT8 code -128 unused.

Figure 3.7 illustrates that quantization can be applied at several different locations. If tensor k contains N_k values represented by b_k bits, the ideal raw storage requirement is

$$M_{\text{raw}} = \sum_{k=1}^K \frac{N_k b_k}{8} + M_{\text{meta}}, \quad (3.29)$$